



UNFCU Lending System Database Connection Test

Diagnostic Tool for Loan Origination System (LOS) Database Connectivity



Current Database Configuration

Server Host:	db-lending-prod-01.unfcu.internal
Port:	5432
Database:	lending_origination_db
Username:	los_app_user
Connection Pool:	min_connections=5, max_connections=20
Timeout:	30 seconds



Connection Status



Waiting for test...

Click "Test Connection (Broken)" to simulate the failure scenario



Diagnostic Tests



Test Connection (Broken)



Apply Fix & Test



Reset Test

Scenario 1: Click "Test Connection (Broken)" to simulate a database connection failure. This will generate multiple failed attempts in the IIS logs.

Scenario 2: After analyzing the logs and identifying the root cause, click "Apply Fix & Test" to correct the connection string and show successful recovery.



Connection Log Output

[INFO] System ready. Click button to start diagnostic test.



Detailed Error Analysis



Performance Metrics

Attempts: 0

Failures: 0

Successes: 0

Avg Time: 0ms

Connection Status

Connection FAILED

Database connection could not be established. Loan origination system is unavailable.

Diagnostic Tests

 Test Connection (Broken)

 Apply Fix & Test

 Reset Test

Scenario 1: Click "Test Connection (Broken)" to simulate a database connection failure. This will generate multiple failed attempts in the IIS logs.

Scenario 2: After analyzing the logs and identifying the root cause, click "Apply Fix & Test" to correct the connection string and show successful recovery.

Connection Log Output

```
[INFO] System ready. Click button to start diagnostic test.
[2026-06-15 02:04:56] ===== STARTING CONNECTION TEST (BROKEN) =====
[2026-06-15 02:04:56] Connecting to: db-lending-prod-01.unfcu.internal:5432
[2026-06-15 02:04:56] Attempting database connection...
[2026-06-15 02:04:57] TCP connection timeout: Unable to reach database server
[2026-06-15 02:04:57] Error Code: TIMEOUT_CONNECTION_REFUSED
[2026-06-15 02:04:58] Retrying connection (attempt 2/3)...
[2026-06-15 02:04:59] Connection refused: No response from server
[2026-06-15 02:04:59] Error Code: ECONNREFUSED
[2026-06-15 02:05:00] Retrying connection (attempt 3/3)...
[2026-06-15 02:05:01] FATAL: Unable to establish database connection after 3 attempts
[2026-06-15 02:05:01] Connection pool exhausted - all connection attempts failed
[2026-06-15 02:05:01] ===== TEST COMPLETED - CONNECTION FAILED =====
```


Detailed Error Analysis

Error Type: Database Connection Timeout
Error Code: ECONNREFUSED
Server: db-lending-prod-01.unfcu.internal
Port: 5432
Status: Connection refused - server not responding

Analysis:

The application attempted to connect to the lending database 3 times, but the server at db-lending-prod-01.unfcu.internal:5432 is not responding. This suggests either:

- X The database server is down
- X The hostname is incorrect or DNS cannot resolve it
- X Network connectivity issue between application and database
- X Database service is not running
- X Firewall blocking the connection on port 5432

Impact:

- X Members cannot submit new loan applications
- X Credit check system unavailable
- X Mortgage processing stopped
- X Balance inquiries failing

Severity: CRITICAL - Revenue impacting outage

Troubleshooting Steps:

1. Check if database server is running: `systemctl status postgresql`
2. Verify network connectivity: `ping db-lending-prod-01.unfcu.internal`
3. Check if port 5432 is open: `netstat -an | grep 5432`
4. Review database logs for connection errors
5. Verify connection string in application configuration
6. Check firewall rules allowing communication

Performance Metrics

Attempts: 3

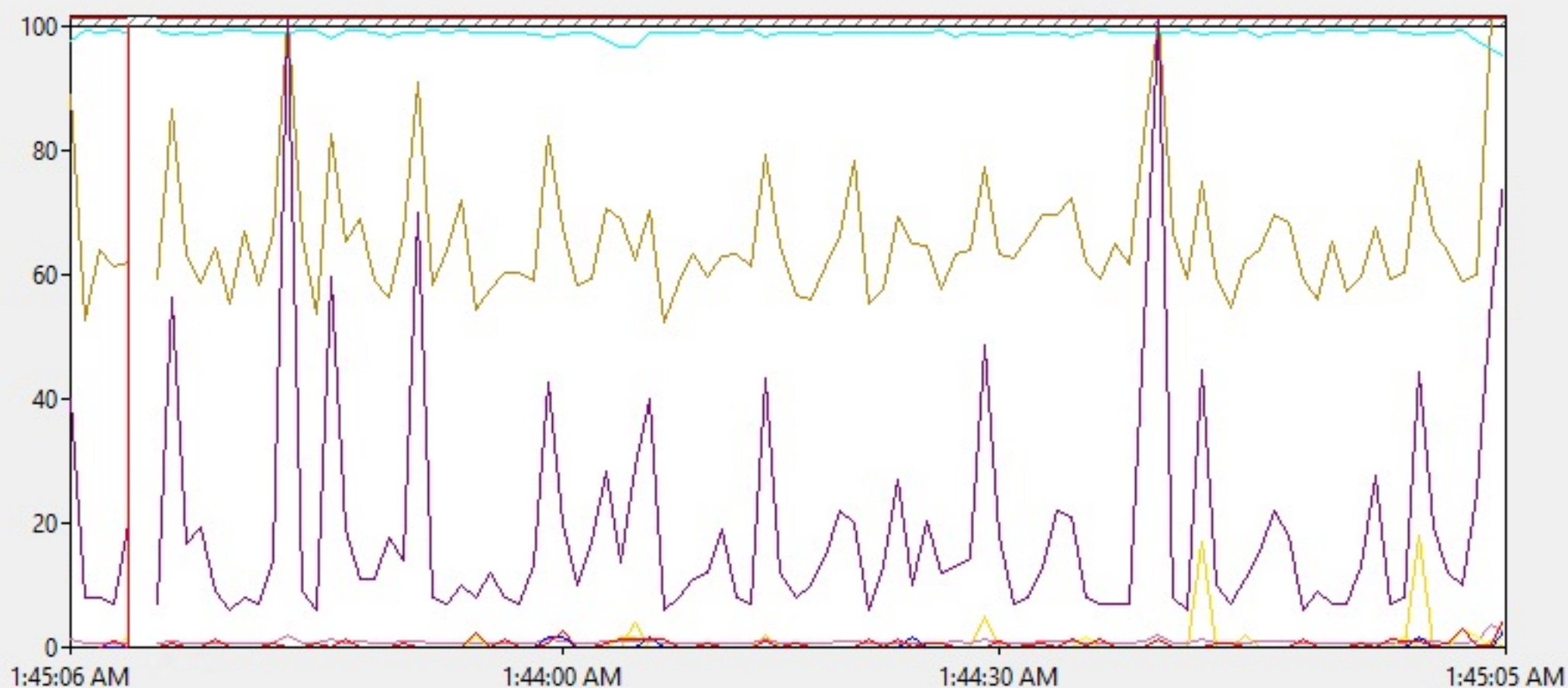
Failures: 3

Successes: 0

Avg Time: 5000ms



- Performance
 - Monitoring Tools
 - Performance Monitor
 - Data Collector Sets
 - Reports



Last 0.165 Average 0.495 Minimum 0.000 Maximum 5.206 Duration 1:40

Show	Color	Scale	Counter	Instance	Parent	Object	Computer
☒	Red	1.0	% Processor Time	_Total	---	Processor Information	\\unfcu-iis-demo
☒	Red	1.0	Available MBytes	---	---	Memory	\\unfcu-iis-demo
☒	Green	1.0	% Processor Time	_Total	---	Processor	\\unfcu-iis-demo
☒	Blue	1.0	% User Time	_Total	---	Processor	\\unfcu-iis-demo
☒	Yellow	1.0	% Privileged Time	_Total	---	Processor	\\unfcu-iis-demo
☒	Pink	0.01	Interrupts/sec	_Total	---	Processor	\\unfcu-iis-demo
☒	Cyan	1.0	% DPC Time	_Total	---	Processor	\\unfcu-iis-demo
☒	Purple	1.0	% Interrupt Time	_Total	---	Processor	\\unfcu-iis-demo